
Nathan Sauldubois

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Summary —

I am a postdoctoral researcher at **New York University (NYU)**, working on probability, optimal transport, martingale optimal transport, and model risk. I obtained my **PhD in Applied Mathematics from École Polytechnique** under the supervision of **Nizar Touzi**, and defended my dissertation on **November 25, 2025**. My research combines theoretical probability, optimal control, optimal transport, and distributionally robust optimization.

Skills

Mathematical Interests: Optimal transport, martingale optimal transport, probability, optimal control, mean-field games, robustness

Numerical Interests: Machine learning, deep learning for PDEs, entropic regularization, particle methods
Programming: Python, R, OCaml, $\text{\LaTeX}$
Languages: French (native), English (fluent)

Publications

Published Papers & Preprints

- Sauldubois, N., and Touzi, N. (2024). *First-Order Martingale Model Risk and Semi-Static Hedging*. arXiv preprint arXiv:2410.06906. Under revision at *SIAM Journal on Financial Mathematics*.
- Compont, A., Sauldubois, N., and Touzi, N. (2025). *Sensitivity Analysis of Distributionally Robust BSDEs and RBSDEs*. arXiv preprint arXiv:2511.01828. Under revision at *Mathematical Finance*.
- Sauldubois, N. (2026). *Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach*. arXiv preprint arXiv:2601.20139.
- Sauldubois, N., and Zhang, X. (2026). *Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport*. arXiv preprint arXiv:2605.30560.

In Preparation

- Sauldubois, N. (2024). *Order- n Expansions in Classical and Causal Distributionally Robust Optimization*.
- Sauldubois, N., & Touzi, N. (2023). *A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem*.
- De March, H., Sauldubois, N., & Touzi, N. (2022). *Optimal Transport Meets Healthcare Optimization*.
- Crowell, R., Sauldubois, N., & Touzi, N. (2025). *Distributionally Robust Optimization in Continuous Time Under Law Constraints*.

Talks

- “A Linear–Quadratic Mean-Field Games Approach to the Marketplace Problem”, Séminaire Doctorant du CMAP (2023)
- “Extended L-Q Mean-Field Game with Common Noise”, Hammamet “Stochastic Control and Risk” (April 2023)
- “Robust Hedging and Distributionally Robust Sensitivity”, PhD Talk, Séminaire Bachelier (March 2024)
- “Model Risk and Semi-Static Hedging”, Bachelier Finance Society Meeting (July 2024)
- “Model Risk and Semi-Static Hedging”, Young Researchers Workshop, Berlin (2024)
- “Robust Static Hedging and Distributionally Robust Sensitivity”, Hammamet Workshop, “Stochastic Control and Games for Risk and Regulation” (October 2024)
- “Adapted Wasserstein Gradient Flow and Projected Langevin Dynamics for Weak Optimal Transport”, Hammamet “Stochastic Control and Risk” (April 2026)
- “Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach”, Hammamet “Stochastic Control and Risk” (April 2026)
- “Model Risk Static-Hedging: A Constrained Distributionally Robust Optimization Approach”, XIII Bachelier World Congress, University of Bologna (July 1, 2026)
- “Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport”, DMV Annual Meeting 2026, University of Konstanz (September 9, 2026)
- “Projected McKean–Vlasov Dynamics for Entropic Weak Optimal Transport”, 10th Eastern Conference on Mathematical Finance, Johns Hopkins University, Baltimore (October 2026)

Teaching

Courses, Exercise Sessions, and Tutorials

- **Option Pricing & Stochastic Calculus** — Graduate course, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Spring and Fall 2026)
- **Financial Computing Recitations** — Graduate recitations, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Fall 2026)
- **Summer Bootcamp** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Summer 2026)

- **Winter Math Bootcamp: From Brain Teasers to Black–Scholes** — Graduate bootcamp, M.S. in Financial Risk Engineering (FRE), NYU Tandon School of Engineering (Winter 2026)
- **Probability Refreshers** – Master X–HEC Data Science for Business (2022–2025)
- **Stochastic Models in Finance** – École Polytechnique (2022–2025)
- **Convex Optimization and Control** – Third-year bachelor’s course (2022–2025)
- **Preparatory Classes (CPGE)** – Oral exam preparation tutor at Lycée du Parc (2019–2020)

Supervision

- Co-supervision of the research internship of **Arthur Compoint** on *continuous-time model risk for stochastic optimization problems* (2025).

Administrative Responsibilities

- Responsible for communications and website maintenance for the **Séminaire Bachelier** (CMAP–CREST)
- Co-organizer of the **Séminaire Doctorant du CMAP**, 2023–2024
- Co-organizer with Xin Zhang of the Invited Session “Optimal Transport in Robust Finance” at the 2026 INFORMS Annual Meeting

Education

École Polytechnique <i>PhD in Applied Mathematics</i>	2022–2025
Defended on November 25, 2025. Supervisor: Nizar Touzi.	
ENS de Lyon <i>M2 Advanced Mathematics</i>	2020–2021
ENS de Lyon <i>M1 Advanced Mathematics</i>	2019–2020
ENS de Lyon <i>L3 Fundamental Mathematics with a Physics Minor</i>	2018–2019

Research Experience

Postdoctoral Researcher, NYU	2025–Present
– Distributionally robust optimization in continuous time. – Adapted Wasserstein gradient flows and applications to entropic martingale optimal transport and weak optimal transport.	
CMAP, École Polytechnique <i>PhD Research</i>	2022–2025
– Mean-field games with control-dependent interactions. – Sensitivity analysis and distributional robustness in stochastic control. – Martingale optimal transport and model risk.	
Natixis – Quantitative Research	2021–2022
<i>Research Internship supervised by Pierre Henry-Labordère</i>	
– Model uncertainty in fixed-income products. – Super-replication strategies for Bermudan swaptions with non-constant notionals. – Quantum optimal transport methods for correlation swaps.	
CMAP, École Polytechnique	2021
<i>Research Internship supervised by Nizar Touzi and Hadrien De March</i>	
– Entropic optimal transport applied to healthcare policy design, in partnership with Quantev.	
Oxford University	2020
<i>Research Internship supervised by Julien Berestycki</i>	
– Analysis of high-dimensional branching Brownian motion and extremal particle estimates.	
Université de Bordeaux	2019
<i>Research Internship supervised by David Lannes</i>	
– Study of shallow-water wave equations and the behavior of solutions in bounded-depth domains.	

Additional Activities

- Special Jury Prize – SMF Junior Contest (2018)